

Duration of Syllable Nuclei in English*

GORDON E. PETERSON AND ILSE LEHISTE

Speech Research Laboratory, University of Michigan, Ann Arbor, Michigan

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This study deals with the influence of preceding and following consonants on the duration of stressed vowels and diphthongs in American English. A set of 1263 CNC words, pronounced in an identical frame by the same speaker, was analyzed spectrographically, and the influences of various classes of consonants on the duration of the nucleus were determined. The residual durational differences are analyzed as intrinsic durational characteristics, associated with each syllable nucleus. The theory is tested with a set of 30 minimal pairs of CNC words, uttered by five different speakers.

INTRODUCTION

IN the investigation of the relation of basic speech parameters to their linguistic interpretation, there appear to be two general approaches which may be taken. In one approach speech synthesis is employed, and the parameters under study can be controlled individually and accurately. The limitation of speech synthesis is that it may be necessary to employ incomplete sets of parameters or parameters which do not correspond closely in their properties to those generated in human speech. In the second approach, actual speech is recorded for analysis, and thus the parameters may be procured in their natural form. The limitation of this method is that it is sometimes difficult to measure the parameters accurately, and it is even more difficult to obtain their precise control. It appears that in the present stage of speech technology, useful information can be derived from both techniques of investigation.

The present study involves the latter method, in which samples of actual speech are recorded and analyzed. This paper reports some of the results of a study of stressed syllable nuclei in English, and deals with certain aspects of duration.

Duration may have different linguistic functions in different languages. In certain languages, a meaningful difference may be associated with a change in the duration of a consonant or vowel. In some languages, however, changes in the duration of a sound may be determined by the linguistic environment and may be associated with preceding or following segmental sounds, initial or final position in an utterance, or type and degree of stress. Such durational changes in turn may become cues for the identification of the associated phonemes. In a recent study of acoustical cues to word boundaries,¹ it was found that allophonic variations of sounds at juncture points included significant changes in duration. In order to identify changes in duration which are cues to juncture, it is necessary to isolate such changes from durational features that form an

integral part of the distinctive properties characterizing the various English phonemes.

It has been observed before that the duration of the vocalic part of a monosyllabic word may be conditioned to some extent by the class of consonants which follow the syllable nucleus. House and Fairbanks,² and more recently Denes,³ have shown this to be the case for English; Zimmerman and Sapon⁴ have examined the problem cross-linguistically and have suggested that such observed variations are a part of the phonetic system of English rather than a general principle applicable to all languages induced by physiological factors.

This paper represents a further examination of some of the characteristics of duration in English. If the factors that condition the duration of syllable nuclei can be isolated and described, and if their influence on the syllable nucleus can be determined, the remaining durational differences, if any, might be defined as intrinsic durational differences associated with each vowel pattern. If such intrinsic durational differences exist, it would be of interest to determine their relation to various types of syllable nuclei, as monophthongs, glides, and diphthongs. Thus durational aspects might contribute to the arrangement of the phonetic data of syllable nuclei into a phonemic pattern.

MATERIALS AND PROCEDURES

Connected speech was used for the experiments. The material studied consisted of two sets of data. The first set involved 1263 words, selected on the basis of frequency of occurrence, and recorded by one speaker in an identical sentence frame, with determined stress and pitch patterns. The words all have the same structure, consisting of one of fifteen common syllable nuclei of English, preceded and followed by a consonant phoneme. The list is described in more detail in a previous publication.⁵ The second set involved 70 minimally

² A. S. House and G. Fairbanks, *J. Acoust. Soc. Am.* **25**, 105-113 (1953).

³ P. Denes, *J. Acoust. Soc. Am.* **27**, 761-764 (1955).

⁴ S. A. Zimmerman and S. M. Sapon, *J. Acoust. Soc. Am.* **30**, 152-153 (1958).

⁵ I. Lehiste and G. E. Peterson, *J. Acoust. Soc. Am.* **31**, 280-286 (1959).

* This research was supported (under contract) by the Information Systems Branch of the Office of Naval Research.

¹ I. Lehiste, "An acoustic-phonetic study of internal open juncture," Rept. No. 2, Speech Research Laboratory, University of Michigan, Ann Arbor, Michigan (August, 1959).

different words, including 60 CNC words forming 30 minimal pairs, and 10 disyllabic words constituting five additional minimal pairs; the 70 words were uttered by five different speakers of the same general dialect. These five speakers used the same frame sentence and pitch and stress pattern as the speaker for the larger set of material. The 1263-word CNC list and the 350 items recorded by five speakers were analyzed acoustically by various techniques. Broad-band and narrow-band spectrograms were made of the total set. Analyses were also made of the total set with the Mingograph at the Royal Institute of Technology in Stockholm. These analyses included two intensity curves with high-frequency and low-frequency pre-emphasis, a Grützmaier pitch curve, and a duplex oscillogram. The smaller set of 350 items was also processed through the 48-channel spectrograph available at Stockholm.⁶ The data measured from the spectrograms form the primary basis for the observations presented in this paper.

SEGMENTATION

An essential problem in the measurement of the duration of syllable nuclei is that of segmentation. Segmentation has long been and continues to be a major problem in speech analysis. Basic difficulties in the concept of segmentation have been discussed previously by one of the present authors.⁷ Since vowel and consonant lengths contrast in some languages there can be no question but that duration may be linguistically distinctive. The specification of those cues which are perceptually significant in linguistic judgments of duration is a subject which should receive further study.

Successive speech sounds not only involve physiological targets and controlled movements, but they often involve changes from one sound type to another. These changes often occur at distances far removed from the targets and, for the purposes of this study, were considered the segment boundaries. Such changes obviously represent major points of transition in the activities of the vocal mechanism. If speech perception is closely correlated with these activities, it seems probable that the transitional regions may be of considerable linguistic significance. The perceptual and linguistic significance of such boundaries merit much further investigation. For purposes of automatic speech recognition, it is obviously necessary to employ some procedure for segmentation or quantization.

There are many instances in which the cues signalling the beginning and the end of a syllable nucleus are relatively unambiguous, but there are many other

instances where it is very difficult to specify the point of segmentation. An attempt will be made to describe the major cues that were used in the segmentation basic to the measurements of the present study. It should be emphasized that the procedures employed in this study sometimes involved a great deal of human judgment. In several instances, segmental cues of a type not anticipated were observed. We are in no position to consider the universality of these cues, but it appears profitable to investigate some of them further.

In the present study, instrumental accuracy is in general considerably greater than the accuracy with which the segmental boundaries can be determined. It was usually possible to determine segmental boundaries within one or two centiseconds. In some instances, however, the transitions between consonants and vowels involve an overlapping of cues, and in such instances it does not appear meaningful to attempt to determine exact time boundaries.

1. Initial and Final Plosives

The release of a voiceless initial plosive appears as a spike on the spectrogram. The duration of the explosion depends on the bandwidth of the major resonance and is followed by a period of frication and a period of aspiration.⁸ Two separate measurements were made for syllable nuclei following aspirated plosives, one from the center of the releasing spike and the other from the onset of voicing immediately after the aspiration. There was usually a measurable concentration of fricative energy in the regions of higher formants throughout the aspiration period, and it was difficult to decide whether at a given moment the pattern in these formants represented breathy phonation or modulated fricative energy. The onset of voicing could be determined relatively accurately, however, by observing the first formant. There was often a weak energy concentration at the frequency of the first formant during the period of aspiration, and the onset of voicing was clearly distinguishable. Thus it was usually possible to determine the frequency of the first formant, both immediately after the release of the plosive and at the onset of voicing after the aspiration. In vowels involving high first formants (particularly /a/, /æ/, and /ɔ/), the energy concentration in the region of the first formant during aspiration was often comparable to that at other formant frequencies, but the onset of voicing was usually clearly distinguishable as the moment in time at which periodic striations started in the first formant frequency.

After voiced initial plosives, the period of aspiration was absent, but the period of frication following the

⁶ The Mingograph, a recording oscillograph with a relatively high-frequency response, and the 48-channel spectrograph are described by C. G. M. Fant, "Modern instruments and methods for acoustic studies of speech," *Proceedings of the VIII International Congress of Linguists* (Oslo University Press, Oslo, 1958).

⁷ G. E. Peterson, *Language* 31, 414-427 (1955).

⁸ The distinction between explosion and frication is a matter of source. Explosion is considered to be the sound produced by the shock excitation of the vocal cavities due to the pressure release, and frication is the sound which originates from turbulence produced by the flow of air through the narrow passage which is formed immediately after the release. Cf. Fant, footnote reference 6, pp. 307-308.

spike was usually more prominent than in the case of voiceless plosives. The measurements were again made from the center of the spike, so that the frication period was included in the duration of the vowel. The duration of the frication varied between about 0.5 and 2.5 csec.

The beginning of final voiceless plosives was determined by the abrupt cessation of all formants. The final voiced plosives were often pronounced with full voicing and a voiced release; under the conditions of recording, a considerable amount of energy was present in the voiced plosives, and up to fifteen harmonics appeared in some of the narrow-band spectrograms. Thus the cessation of voicing was not a proper cue for the termination of the syllable nucleus. Instead, the beginning of final voiced plosives was determined by comparing narrow-band and broad-band spectrograms, from which the moment in time when the energy in the higher harmonics was suddenly greatly diminished could be ascertained. In general, it was possible to specify the boundary with an accuracy of about one vocal-fold period.

Examples of initial and final plosives may be seen on Figs. 1-4. Figures 3 and 4 illustrate initial voiced and voiceless plosives; Figs. 1 and 2 contain examples

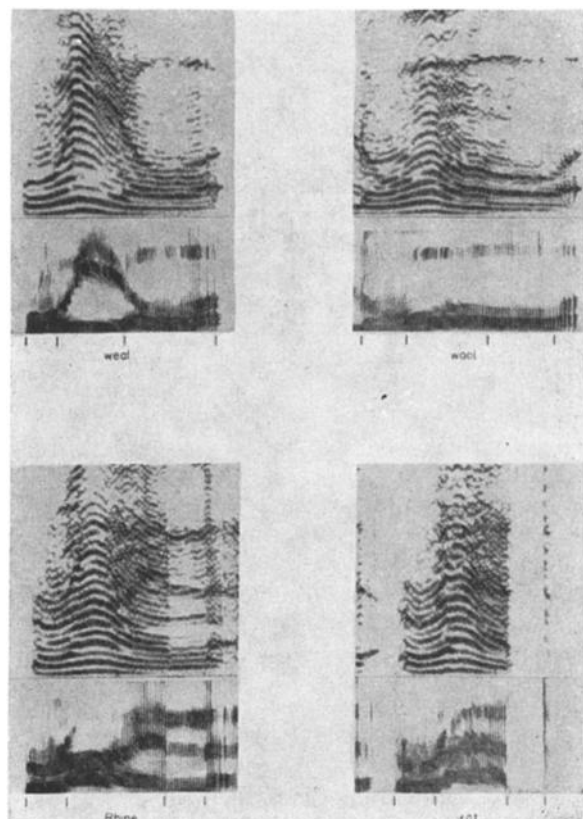


FIG. 1. Broad-band and narrow-band spectrograms of four CNC words spoken by informant GEP. Approximate segmentation points for identifying the boundaries of initial and final consonants have been provided.

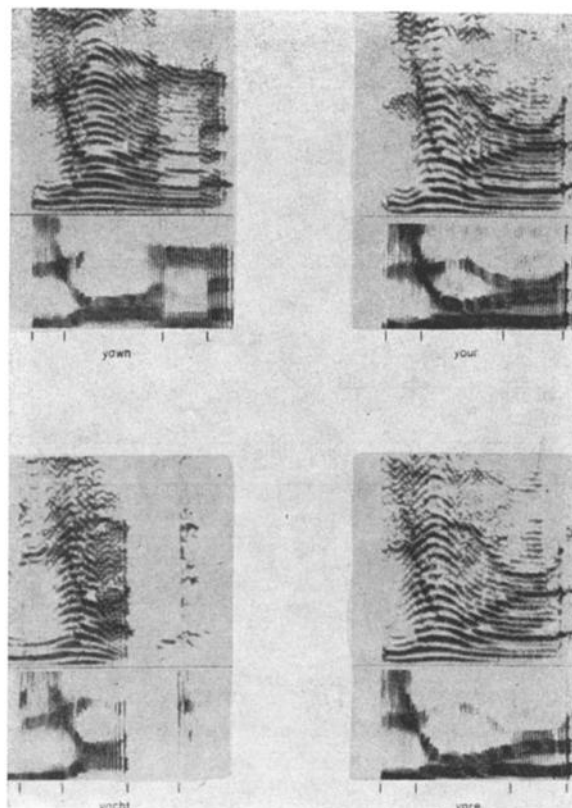


FIG. 2. Broad-band and narrow-band spectrograms of four CNC words spoken by informant GEP. Approximate segmentation points for identifying the boundaries of initial and final consonants have been provided.

of final voiceless plosives, and Fig. 4 contains examples of both voiced and voiceless final plosives.

2. Initial and Final Nasals

In the measurement of syllable nuclei durations, initial nasals offered no difficulty. It was usually possible to identify the vocal-fold period which followed the velar closure by observing the abrupt change from steady formant pattern to rapid onglide movement. Final nasals share the characteristic of steady resonances with initials. In the case of two speakers from the smaller set of data, however, the vowels were nasalized considerably. This had no significant effect upon the identification of the initial boundary of the syllable nucleus, but for these two speakers the nasalization of the vowels obscured the transition from the syllable nucleus to the final nasal consonant on the broad-band spectrograms. The control set of 70 words contained 14 final nasals, but only 3 initial nasals; the relative ease of identifying the boundaries of initial nasals may be due to the very limited data. In the narrow-band spectrograms for these speakers, it was possible to locate the approximate boundaries as the position at which there was a sudden change in the relative marking of the various harmonics. Those

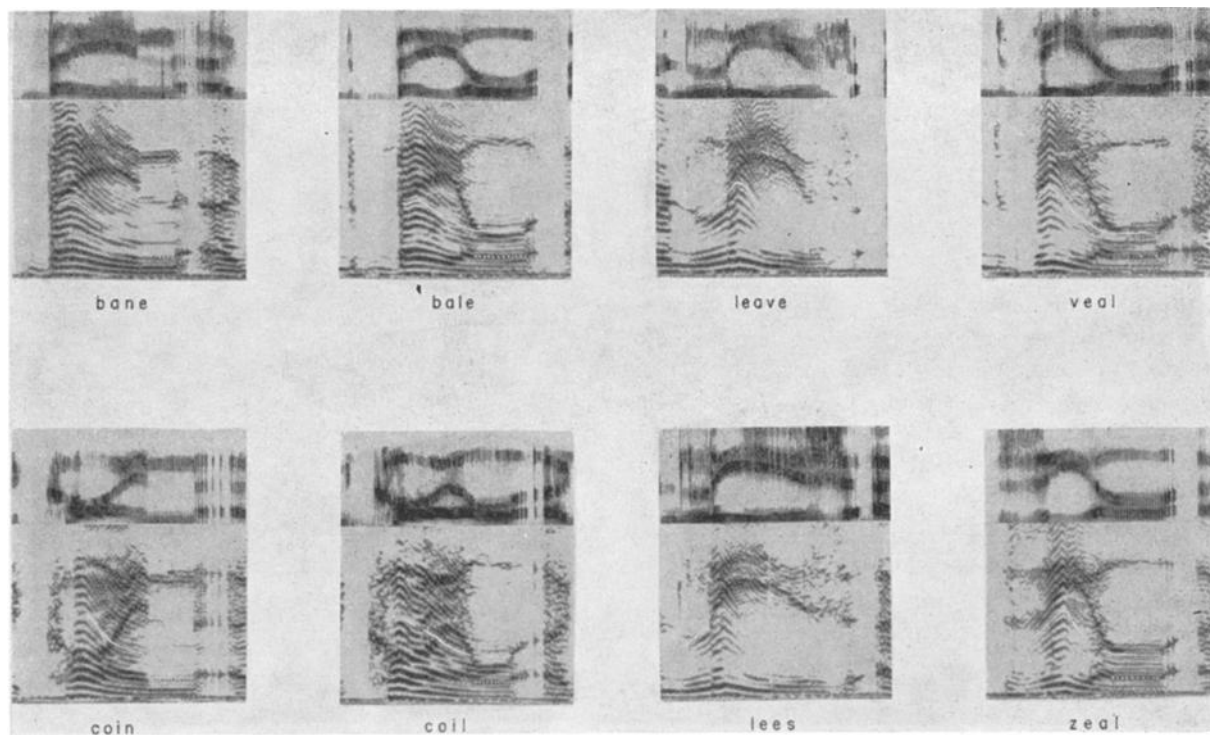


FIG. 3. Broad-band and narrow-band spectrograms of eight CNC words uttered by informant GEP. The tenth harmonic has been traced in white paint on the narrow-band spectrograms; the fifth harmonic has been traced with a dotted white line on some of the spectrograms.

harmonics that were not within the frequency region of either the oral or nasal resonances were marked much more lightly following the vocalic period, and thus a boundary point could be established. Information on relative changes in energy level among the vowel and nasal formants and the minima (or valleys) between is not yet available. Thus the extent to which the marking changes are due to energy-level changes cannot be specified at present. The pattern changes result in part, of course, from the automatic adjustments of the

narrow-band marking control to the decrease in over-all output energy during the formation of nasals.

Some examples of final nasals may be found on Figs. 1, 2, and 3. The narrow-band patterns in Figs. 1 and 2 are particularly good illustrations of the changes in marking of the harmonics outside the resonances.

3. Initial and Final Fricatives

The beginning of a vowel after an initial voiceless fricative was determined by the onset of voicing in the region of the first formant. This cue was also employed in determining the beginning of the syllable nucleus after an initial /h/, as formant movements were not adequate indications of the points of transition. In such cases, the intensity curves provided a valuable additional reference. There was a period of "breathy" quality for initial /h/ on broad-band spectrograms after the onset of voicing (noise pattern superimposed upon a rather clear formant pattern), but the intensity curves provided a relatively unambiguous cue. Some of the initial voiceless fricatives registered considerable energy on both Mingograph traces and the oscillogram, but there was a cessation of fricative energy before the onset of phonation, and a sharp minimum in the intensity curves provided an appropriate boundary point.

The terminal boundaries of initial voiced fricatives were, in general, rather easily recognized on broad-band spectrograms. The superimposed noise usually ended

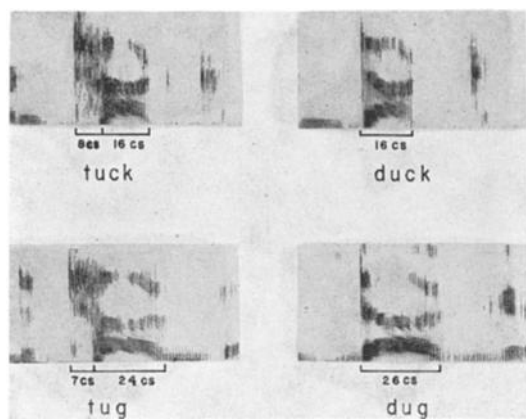


FIG. 4. Broad-band spectrograms of four CNC words spoken by informant GEP. The duration of the aspiration following a voiceless initial plosive has been presented separately from the duration of the voiced part of the syllable nucleus.

abruptly. Final voiceless fricatives were recognized by the onset of random noise: The vowel was considered terminated at the point where the noise pattern began, even though voicing in a few low harmonics continued for a few centiseconds in most cases. Final voiced fricatives were more troublesome. In broad-band and narrow-band spectrograms, the transition between vowel and consonant appeared rather gradual, but the onset of high-frequency energy in the case of /z/ and /ʒ/ provided a clear boundary on the intensity curves. The boundaries preceding final /v/ and /ð/ were recognized chiefly by the rapid decrease of energy that could usually be detected on the intensity curves.

Figure 5 presents 4-channel Mingograph tracings of two of the frame utterances spoken by informant Br. Curves *a* and *b* display the sound intensity; low-frequency pre-emphasis was employed in constructing curve *a* and high-frequency pre-emphasis was employed for curve *b*; *c* is a fundamental frequency contour, and *d* a duplex oscillogram.⁶ The top utterance represents the sentence "Say the word 'voice' again," and may be compared with "Say the word 'noise' again," presented in the lower half of the illustration. The sharp boundary between initial /s/ and the following vowel in *say* can be observed in both utterances. The boundaries of the voiceless sibilant /s/ in *voice* are more clearly demarcated than the initial boundary of the /z/ in *noise*; curve *b*, which emphasizes energy in the higher frequencies, provides the best clue for isolating final /z/. The initial voiced fricative /v/ in *voice* can be best isolated on curve *a*, with low-frequency pre-emphasis. Initial and final voiced fricatives may also be observed on Fig. 3, which is a good illustration of the difficulty of finding a clear-cut boundary line for final voiced fricatives if only broad-band and narrow-band spectrograms are used for the analysis.

4. Initial /w/ and /y/

Both of these initial consonants involved a steady-state period. Since they were fully voiced and had only a minimal amount of friction, the formant movements from the consonant to the syllable nucleus were uninterrupted. Nevertheless, certain cues appeared with a fair amount of regularity, and together provided reasonably usable criteria for segmentation. For initial /w/ the region in which the slope of the second formant acquired a positive value was considered the boundary. This directional change was often accompanied by a sharp increase in energy, and the energy change was accompanied on the narrow-band spectrogram by a darker marking of the harmonics not in the frequency regions of resonance. Such energy cues were particularly useful in sequences where /w/ was followed by a vowel with a low second formant. Figure 1 contains initial /w/ followed by a front and a back vowel. A steady state for /w/ is followed by a rather sharp upward inflection of the second formant in the sequence /wi/;

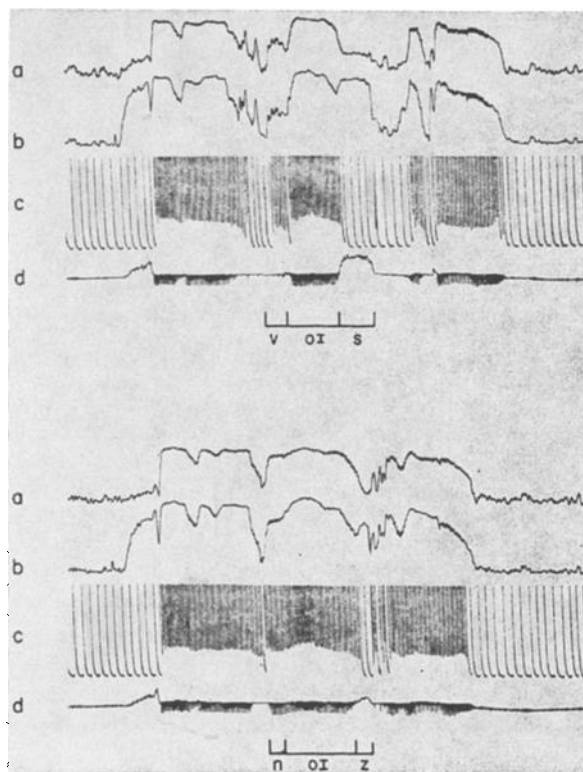


FIG. 5. Four-channel Mingograph tracings of two utterances by informant Br: "Say the word 'voice' again," and "Say the word 'noise' again." Curve *a* is an intensity curve with low-frequency pre-emphasis; curve *b* is an intensity curve with high-frequency pre-emphasis; curve *c* is a pitch curve (modified Grützmacher method); and curve *d* is a duplex oscillogram. The analysis was performed at the Royal Institute of Technology in Stockholm.

there is a noticeable change in the marking of the formants and also in the harmonics not within the resonance regions in the case of the sequence /wu/.

In these data, initial /y/ had a steady state, in which the frequency of the third formant was much greater than that for any simple vowel nucleus. The point, however, at which the transition to the following vowel began was at a considerably lower position in frequency. Thus the third formant of /y/ performs a rapid dip in frequency before rising back to the third-formant position of the vowel. The position in time of the frequency minimum of the third formant was thus considered the point of onset of the following vowel. This cue is most easily determined for vowels with a high third-formant frequency. Figure 2 contains illustrations of initial /y/. The steady state associated with the initial /y/ is characterized by a high third formant and a relatively weak intensity of the harmonics between the resonances; the minimum in the movement of the third formant from the steady state of /y/ to the following vowel serves as the point of segmentation.

5. Initial and Final /l/ and /r/

Both initial /l/ and /r/ had periods of steady resonances. The onset of a vowel after /l/ was usually un-

ambiguously defined on the narrow-band spectrograms by the sudden change in marking of the harmonics between the various resonances at the change from steady formants to onglide. Initial /r/ often had a slight fricative quality. In addition, the frequency movements of the third formants usually provided a clue for the segmentation.

Final /l/ and /r/ presented particularly difficult problems. Very often the formant movements were quite smooth, and the establishment of a boundary on the basis of broad-band spectrograms was questionable. Intensity curves were helpful in instances where the vowel had an intrinsic energy considerably different from that of /l/ or /r/. In the transition from the vowel nucleus to /l/ a frequency minimum or a relatively rapid rise in the frequency of the third formant was sometimes present and was used as the basis of the segmentation. In the total set of data, the third formant of /l/ had an average frequency value of 2635 cps, and the change from the usually lower third-formant position of the vowel to that of /l/ sometimes involved a well-defined change point. But in a rather large number of instances, the formant movements appeared smooth, there was no significant change in the intensity, and the determination of a boundary had to be accomplished by some other criterion.

It was observed that the fundamental voice-frequency curve employed in each utterance had certain characteristic distributional features. The initial consonant of the word in the frame appeared to determine the occurrence of the peak of the fundamental curve. When the initial consonant was voiced, the peak occurred in the middle of the nucleus of the target word, with a rather smooth glide of the fundamental (usually upward but often down on voiced plosives) during the initial voiced consonant. If the consonant was voiceless, and particularly when the consonant was a voiceless fricative, the peak occurred immediately at the onset of voicing, and the fundamental on the syllable nucleus thereafter decreased. The total drop in pitch normally took place during the vocalic part of the syllable. When the final consonant was voiceless, this might have been expected. But when the final consonant was voiced, the fundamental pitch reached its minimum value by the beginning of the consonant and remained almost completely level for the duration of the consonant. This pattern was observed with very great regularity, and we concluded that in this type of utterance the region where the fundamental has become essentially level may be considered the consonant part.

Figure 3 illustrates the parallelism between such words as *coin* and *coil*, where the fundamental voice frequency was used to determine the boundary between the syllable nucleus and final /l/. The tenth harmonic has been traced in white, and provides a visual representation of the fundamental movement associated with this intonation. Some examples of initial /l/ are also included in Fig. 3 to illustrate that in initial position,

the rise in pitch takes place during the pronunciation of the voiced resonant; level fundamental frequency is not a characteristic of all /l/ sounds.

Examples of final /r/ may be found on Fig. 2, where the segmentation has been based on two clues: the steady fundamental frequency associated with final position, and the change in the third-formant frequency. Initial /r/ sounds are represented in Fig. 1. Approximate segmentation may be achieved by comparing the relative markings of the harmonics in the narrow-band spectrogram and by identifying the position at which the third formant begins to rise rapidly in frequency in the broad-band spectrogram.

PROBLEM OF TEMPO

In order to observe the intrinsic duration of syllable nuclei and the influence of preceding and following segmental sounds upon the duration of syllable nuclei, variability in duration due to the suprasegmental features of pitch, stress, and tempo should be eliminated. Since the recorded utterances involved a uniform stress and intonation pattern, it seems reasonable to assume that stress and pitch affected each test word in essentially the same manner. The chief remaining variable to be considered is speaker rate.

There are several different ways in which speaker rate might be defined for such a set of material. For example, speaker rate might be considered constant if the durations of the sentence frames remained constant. In order to study the durations of the sentence frames, the duration of each test word was subtracted from the duration of the entire utterance. It was found that in both sets of data a glottal stop was often inserted following the test word, and the duration of this stop was included in the measurement of the duration of the test frame. The average durations in centiseconds of the frames for the 70 utterances by each of the five speakers were: 122, 130, 144, 150, and 177. Ten samples of 50 sequential frame sentences each were also extracted from the 1263 words of the larger set of data. The average duration of the frame was 174 csec, with a standard deviation of 6.9 csec. The mean for the five speakers was 144, which differs from the average for the speaker of the large set of data by 30 csec. The average durations of the syllable nuclei contained in the test words in the smaller set of data were: 24, 25, 26, 30, and 31 csec, with an average of 27 csec. The average of the nuclei in the same words for GEP was 28 csec. Thus there was much greater relative variation in the average duration of the frame than in the average duration of the stressed syllable nuclei contained in the test words.

The foregoing data imply that differences in tempo may affect stressed and unstressed words in different ways. It has been suggested previously⁹ that in English

⁹ A. Classe, *The Rhythm of English Prose* (Basil Blackwell, Oxford, 1939).

there is a tendency for stress groups to assume approximately the same duration. Such a principle would require that the length of syllables vary, particularly in the case of stress groups containing different numbers of syllables. Thus we might consider whether the speakers spaced the main stresses at regular intervals. If so, the differences in the durations of stressed test words might be compensated by varying the durations of unstressed syllables.

The durations of the intervals between successive test words were measured for the same set of utterances that was used for computing the average duration of the frames. The selection of a time mark, either within the frame or within the test word, is somewhat arbitrary. If the rate of uttering the unstressed syllables of the frame varied according to the duration of the stressed test words, then a recurrent point in the test frames would not provide a satisfactory measure. On the other hand, since the test words all differ, it is obviously difficult to specify a common point within the test words. The selection of the point of maximum stress within the syllable nucleus might seem most appropriate, but it is undoubtedly influenced by both fundamental voice frequency and intensity in a way which is not yet understood, and the peaks for the two are not necessarily concurrent within the stressed syllable. Since the point of release of the initial consonant of the stressed syllable may be physiologically significant, and since it is a point which can be specified fairly accurately, it was chosen as the time mark for the measurements. The duration of the intervals between successive test words was measured accordingly for the same set of utterances that was used for computing the average durations of the frames.

Figure 6 presents a sequence chart for the durations of the time intervals between successive initial consonant releases of the test words (i.e., the total intervals), the durations of the test words, and the durations of the syllable nuclei. In the graph, relatively long pauses for inhalations, irregular breathing patterns, and relatively uniform rate accelerations and decelerations are all evident.

Figure 7(A) shows the duration of the test word versus the total time interval duration, and (B) shows the duration of the syllable nucleus vs the total time interval duration for 100 utterances by GEP. None of these graphs implies a strong relation between the interval duration measure which we chose and the duration of the test word or the syllable nucleus.

In general, the test word represents only a small fraction of the total duration of the interval. Also, the variation in the interval was small relative to the total interval duration, of the order of 3-5%, or approximately 5 to 15 csec. These facts suggest that variations in rate of utterance by any given speaker had little effect upon the duration of the stressed syllables under study.

A further brief experiment was devised to explore

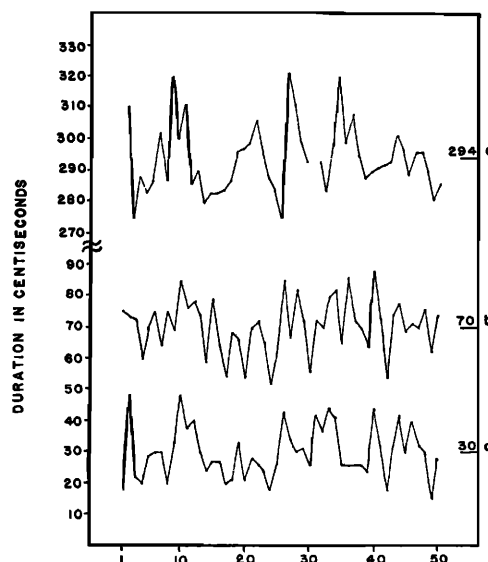


FIG. 6. Utterances 1-50 by informant He. Curve *a* presents the durations of the syllable nuclei, curve *b* the durations of the test words (occurring in random order), and curve *c* the duration of the time intervals between successive initial consonant releases of the test words. The break in curve *c* between items 30 and 31 represents the turning of the page from which the randomized test words were read.

whether stressed and unstressed syllables are similarly affected by a controlled change in tempo. A selected set of test utterances was recorded by three of the speakers. In one instance the spacing in time between the stressed syllables was controlled to be 2 seconds by asking the subjects to speak in synchronism with a periodic pulse produced over an earphone; in a second similar set of recordings, the spacing between stressed syllables was 1 second. The pulse and the speech were recorded on an SPL recorder and on magnetic tape. The durations of the words spoken at both rates of speech were measured, and the speedup factors were computed. In this experiment, the duration of the syllables with main stress changed less than the duration of the unstressed syllables: When the rate of utterance was increased by a factor of two, the stressed words decreased in duration by a factor of approximately 1.5. Under these conditions the actual durations for the stressed words were very similar for the three speakers.

Further investigations of the relations of tempo and stress to duration appear necessary before this variable can be specified adequately. For the present study, however, it seems reasonable to assume that the duration measurements of the test words were not greatly influenced by variations in tempo during the recording by any given speaker.

INFLUENCE OF PRECEDING AND FOLLOWING CONSONANTS

If the influence of suprasegmental factors is eliminated, the remaining durational differences may be

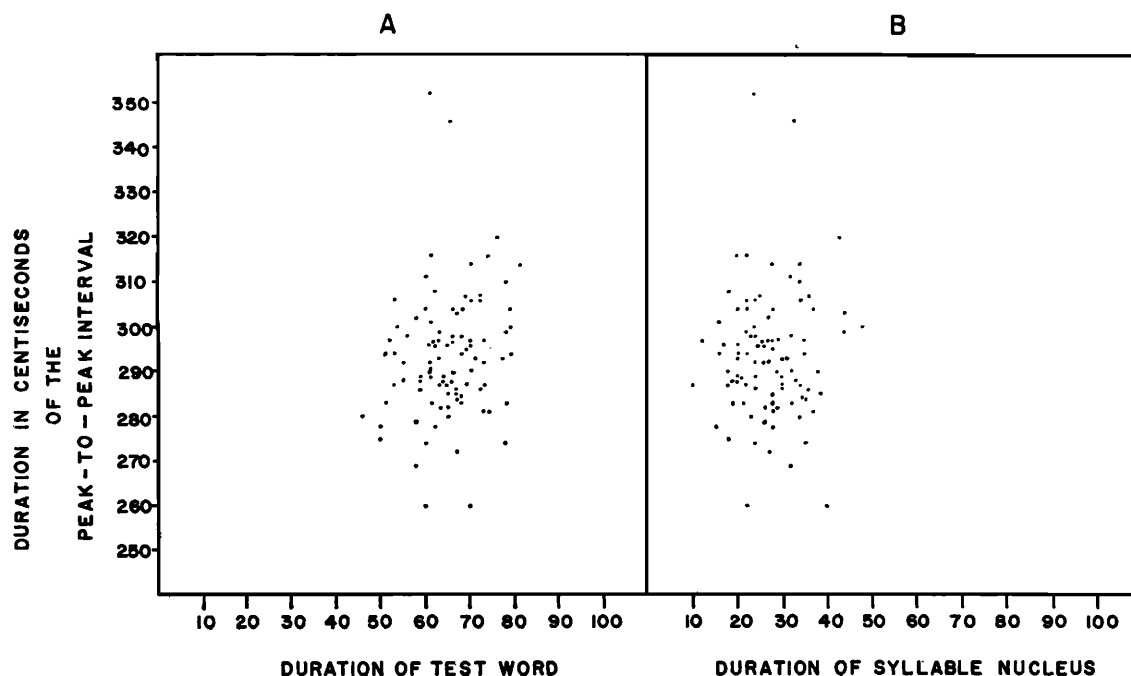


FIG. 7. Durational relationships between test word, syllable nucleus, and total time interval for 100 utterances by GEP. (A) shows the total time interval duration vs the duration of the test word, and (B) shows the total time interval duration vs the duration of the syllable nucleus.

due either to the influence of preceding and/or following consonants, or to intrinsic durational differences. The influence of consonants was considered next. Some of the syllable nuclei are represented by a larger number of words in the CNC list than others. Since we assumed that vowels may have different intrinsic durations, all occurrences of a particular consonant could not be included, as the results would then be weighted according to the predominance of intrinsically short or long syllable nuclei associated with the consonant. Therefore, several smaller subsets, consisting of minimal pairs present in the list, were analyzed. For example, in determining the influence of a following voiced or voiceless consonant on the duration of the preceding syllable nucleus, the durations of the syllable nuclei in such pairs as *beat-bead*, *sight-side*, etc., were compared. There were 118 minimal pairs in the larger set of data, differing in the voicing of the final consonant. The average duration of the syllable nucleus before the voiceless member of the consonant pair was 19.7 csec, and before the voiced member was 29.7 csec. The comparable values for the control group of five speakers were 19.3 csec and 29.1 csec. Thus, for this set of data, the ratio of vowel before voiceless consonant to vowel before voiced consonant is approximately 2:3.

The homorganic nasals influence the preceding vowel in much the same manner as the voiced stops. In 46 instances of triple contrasts, such as *back-bag-bang*, the average values of the syllable nuclei before the voiced stop and nasal were 29.8 csec for the voiced stop and 30.2 for the nasal in the larger set of recordings, and

28.9 csec and 30.0 csec for the stop and nasal, respectively, for the five speakers.

The voiced fricatives appear to have a further lengthening effect. Of the 21 minimal pairs in the set in which the effect of a final voiced plosive could be compared with the effect of the final homorganic voiced fricative, the duration of the syllable nucleus before the voiced plosive was 30.0 csec, whereas the comparable duration before the voiced fricative was 37.9 csec. There were nine sets where four contrasts were represented—sets of such words as *rice-rise-ride-right*. The average durations of the syllable nuclei, in ascending order, were: 18.4 csec before the voiceless plosive, 22.8 csec before the voiceless fricative, 28.0 csec before the voiced plosive, and 37.6 csec before the voiced fricative.

Initial voiced-voiceless contrasts presented no easily discernible pattern. There were 68 minimal pairs in the larger set of data which differed in the voicing of the initial consonant. Figure 8 presents graphically the results of the measurements on these minimal pairs. It appears that there is practically no durational difference associated with an initial /f/-/v/ contrast; in the case of /s/-/z/ and /č/[tʃ]-/j/[dʒ], the effect of voicing appears contradictory, the voiceless member of the pair being followed by a longer syllable nucleus in the case of the /s/-/z/ contrast, but by a shorter syllable nucleus in the case of an initial /č/-/j/ contrast. Initial plosives present a special problem. If aspiration is considered part of the syllable nucleus, then voiceless plosives are regularly associated with longer syllable nuclei than voiced plosives; if, however,

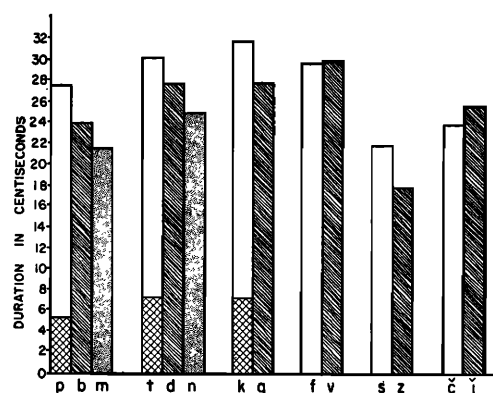


FIG. 8. The duration of syllable nuclei, measured from 68 minimal pairs differing in the voicing of the initial consonant, presented as a function of the initial consonant. The cross-hatched area associated with the columns representing the duration of syllable nuclei following voiceless plosives indicates the duration of aspiration.

the duration of the aspiration is discounted, the syllable nucleus following a voiceless plosive is usually shorter than that following a voiced plosive. In the phonemic analysis of English, voiceless aspirated plosives are commonly analyzed as single phonemes, although they consist of a sequence of acoustical segments.

The aspiration has also been considered as part of the following vowel.¹⁰ According to this opinion, the voiceless part represents the influence of the voiceless consonant as partial progressive assimilation which extends over a certain portion of the duration of the vowel.

The average duration of the aspiration after an initial [p] was 5.8 csec (for 81 different items); after an initial [t] the aspiration lasted 6.9 csec (in 73 instances), and for initial [k] the duration of the aspiration lasted 7.5 csec (for 83 items). These data suggest that aspiration may become progressively longer as the point of articulation shifts farther back in the mouth, but this observation was not supported by a separate analysis of the two main allophones of /k/, [c] and [k]. The average for the front allophone, [c], followed by front vowels (39 instances), was 7.8 csec, whereas the average for the velar allophone [k], followed by back vowels (44 instances), was 7.2 csec. Nevertheless, the /t/ aspirations were consistently shorter than the aspirations associated with either of the allophones of /k/.

If differences in the duration of the aspiration are conditioned and, therefore, nonsignificant, the question could be raised as to whether the duration of the aspiration is irrelevant to the duration of the syllable nucleus. If it is, then the aspiration may be considered as part of the consonant rather than the vowel. In some instances, this actually seemed to be the case.

¹⁰ E. Fişcher-Jørgensen, *Misc. Phonetica* 2, 42-59 (1954).

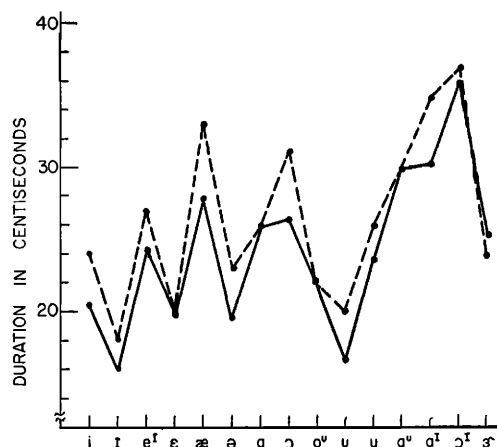


FIG. 9. Average durations of syllable nuclei, measured from minimal pairs differing in the voicing of the final consonant. The solid curve is for a large set of CNC words spoken by GEP; the dotted curve represents the values for 30 minimal pairs uttered by five speakers.

Figure 4 shows such an instance. In the spectrogram for *duck*, the vowel seems as long as the vowel in *tuck*; if the duration of the aspiration is included in the duration of the syllable nucleus, the total duration is more like the duration of the vowel in *dug*.

If we exclude aspiration, the average duration of the syllable nucleus after a voiceless consonant was 25.1 csec; when aspiration is included, it was 30.8 csec; the average duration after voiced consonants was 27.4 csec. These figures represent the average values for all 68 minimal pairs. Perhaps the most obvious conclusion is that, in our data, the influence of an initial consonant on the duration of the syllable nucleus followed no simple regular pattern.

INTRINSIC DURATIONS OF SYLLABLE NUCLEI IN ENGLISH

If the duration of a syllable nucleus is conditioned by the presence or absence of voicing in the following consonant, the intrinsic duration of the syllable nucleus might be defined as the average duration of the respective minimally contrastive pairs. Such average durations were computed for both the large and the small sets of data. In Fig. 9, the solid curve represents the averages for one speaker, and the dashed curve the averages for five speakers. Table I presents the actual values upon which the graph in Fig. 9 is based. The third row adds, for purposes of comparison, the average durations of the syllable nuclei in all instances in which the syllable nucleus occurred in the CNC list before those final consonants that were involved in the respective minimal pairs. In most instances, there is good agreement between the values computed from minimal pairs and the values averaged from a larger number of occurrences. There is also a considerable similarity between the two curves in Fig. 9.

TABLE I. Intrinsic durations of syllable nuclei in American English.

Syllable nucleus	Average for five speakers (in csec)	Average for GEP in minimal pairs	Average for all occurrences in CNC list
i	24	20.6	20.7
ɪ	18	16.0	16.1
eɪ	27	24.3	20.0
ɛ	20	20.3	20.4
æ	33	28.0	28.4
ə	23	19.3	18.1
ɑ	26	26.1	26.5
ɔ	31	26.5	25.0
oʊ	22	22.0	22.2
u	20	16.3	16.3
ʊ	26	23.8	23.5
aʊ	30	30.2	30.2
aɪ	35	30.3	31.0
ɔɪ	37	36.0	36.0
r	24	25.3	25.6

It appears rather clear that the vocalic syllable nuclei may be subdivided into classes according to their durations. As a first approximation, the syllable nuclei may be considered as consisting of four short nuclei, [ɪ], [ɛ], [ʊ], and [ə], and nine long syllable nuclei.

The average duration of the nuclei preceding each consonant was next computed, and Fig. 10 presents these durations as a function of the final consonant phoneme. Only minimal pairs were included in constructing the two curves of Fig. 10 and in the accompanying Table II. The top curve is for all long syllable nuclei, and the bottom curve is for the four intrinsically short syllable nuclei. The voiceless-voiced contrasting influence on syllable nucleus duration is clearly evident in the variations of both curves. It is interesting that

TABLE II. Duration of syllable nuclei as a function of the following consonant.

Following consonant	Duration of short syllable nucleus (in csec)	Duration of long syllable nucleus (in csec)
p	13.8	18.8
b	20.3	30.7
t	14.7	21.0
d	20.6	31.8
k	14.5	20.0
g	24.3	31.4
m	22.0	31.3
n	21.6	32.2
ŋ	21.8	35.0
f	19.2	26.1
v	23.1	37.4
θ	20.8	26.5
ð	26.0	38.1
s	19.9	26.9
z	26.2	39.0
ʃ	21.2	27.8
ʒ	...	41.0
r	22.6	29.6
l	21.8	29.3
ç	14.5	19.8
j	19.1	30.0

there is durational overlap between the durations of long nucleus plus voiceless consonant and short nucleus plus voiced consonant, for example in such sets as *bead-beat-bit-bid*. A further interesting fact is that the affricates [tʃ] and [dʒ] affect the preceding vowel durations in the same manner as the plosives. There is a possible historical basis for this similarity. Other characteristics of the two durational classes of syllable nuclei will be discussed in a subsequent issue.

CONCLUSION

It appears from the data analyzed during the present study that the durations of all syllable nuclei in English are significantly affected by the nature of the consonants that follow the syllable nuclei; the influence of the initial consonants upon the durations of the syllable nuclei appears to be negligible. In general, the

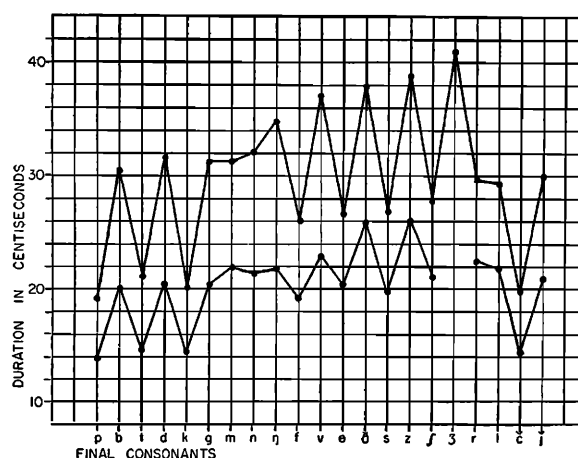


FIG. 10. Duration of syllable nuclei as a function of the final consonant phoneme, computed from minimal pairs of CNC words spoken by GEP. The top curve presents the average durations for all long-syllable nuclei and the bottom curve the averages for four short-syllable nuclei.

syllable nucleus is shorter when followed by a voiceless consonant, and longer when followed by a voiced consonant. In a large number of minimal pairs of CNC words differing in the voicing of the final consonant, the ratio of the durations of the vowels was approximately 2:3, the syllable nucleus before the voiced consonant being longer in every case. As a class, plosives are preceded by the shortest syllable nuclei; nasals had approximately the same influence as voiced plosives. Syllable nuclei were longest before voiced fricatives. Assuming that a voiced consonant has a lengthening influence on a syllable nucleus that is comparable to the shortening influence of a voiceless consonant, the intrinsic duration of syllable nuclei may be defined as the average duration of the syllable nucleus measured from minimal pairs differing in the voicing of the final

consonant. Such intrinsic durations were computed for all stressed syllable nuclei in English. Two groups may be established according to this procedure: intrinsically short syllable nuclei, comprising the four vowels [i, e, ə, u], and intrinsically long syllable nuclei, consisting of [ɪ, e^I, æ, ɑ, ɔ, o^U, u, r, ɑ^U, ɔ^I].

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Ultrasonic Studies in Melts and Solutions

S. V. SUBRAHMANYAM AND J. BHIMASENACHAR

Department of Physics, Sri Venkateswara University, Tirupati, India

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The variation of ultrasonic velocity with concentration of phenyl salicylate, o-chloronitrobenzene, stearic acid, naphthalene, o-nitrophenol, m-nitrophenol, and palmitic acid in suitable solvents has been studied using an ultrasonic interferometer. The linear plots of velocity vs percentage weight of the solute are extrapolated to evaluate the velocities corresponding to 100% of the solute in each case. The agreement between the values obtained from solutions using different solvents is good. Sound velocities in the melts of the above substances have also been determined. It is found that the characteristic velocity obtained from data on solutions corresponds to the velocity at melting point in all cases except in that of naphthalene.

INTRODUCTION

INVESTIGATIONS regarding the variation of ultrasonic velocity with concentration of a solute in different solvents lead to interesting data characteristic of the solute. It is known^{1,2} that the plots of velocity vs percentage weight of solute in solution are linear for a large number of solute-solvent systems. It is possible that there may be solute-solvent systems for which the velocity vs percentage weight plots are not strictly linear. However, for those cases for which the plot is linear, the velocity for 100% of the solute can be evaluated by extrapolation of the graph. The extrapolated value of the velocity for a solute is found to be independent of the nature of the solvents used. The velocity so obtained should then be a characteristic of the solute. In the present investigation, it is proposed to examine to what phase (solid or liquid) of the solute this characteristic velocity refers.

Ultrasonic velocities of phenyl salicylate, o-chloronitrobenzene, stearic acid, naphthalene, o-nitrophenol, m-nitrophenol, and palmitic acid have been studied in solution and molten conditions. In the case of naphthalene, velocity of longitudinal waves in the solid state has also been determined.

EXPERIMENTAL DETAILS

(A) Ultrasonic Interferometer

A single crystal interferometer has been employed to determine the ultrasonic velocities in solutions and melts. An x-cut quartz of natural frequency 750 kc is

excited by a precision variable frequency oscillator. Impedance variations of the quartz crystal, caused by resonance in the liquid column at half-wavelength intervals, are detected by an rf detector. The distance traveled by the reflector for 20 half-wavelengths is measured. A number of readings are taken over different portions of the interferometer screw. Ultrasonic velocities are calculated using the average values of such measurements, the differences between readings taken at different portions of the screw being small. It is estimated that the determinations of the velocities are subject to an error of $\pm 0.15\%$.

Solutions are prepared by dissolving weighed quantities of A.R. quality solutes in known volumes of A.R. quality solvents. Densities of such solutions are determined by the hydrostatic method. All the measurements on solutions are made at $30^\circ \pm 0.5^\circ\text{C}$.

For the determination of ultrasonic velocity in melts, an electrical furnace is built around the interferometer cup, which contains the substance. The temperature of the furnace is maintained within $\pm 0.5^\circ\text{C}$ by suitably controlling the current passing through it. Measurements are taken after allowing the melt to stand for half an hour at each temperature. Consistency of the distance traveled by the reflector for a particular number of half-wavelengths over different regions is taken as an indication of the nonexistence of temperature gradient in the liquid column.

(B) Composite Piezoelectric Oscillator

To determine the ultrasonic velocity in solid naphthalene, the composite piezoelectric oscillator³ method is

¹ Schaffs, Z. Physik. **105**, 658 (1937).

² Sibiya and Narasimhaiya, J. Mysore Univ. **1**, 133 (1941).

³ S. V. Subrahmanyam, J. Chem. Phys. **22**, 1562 (1954).